Public Option and Private Screening

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Research Question

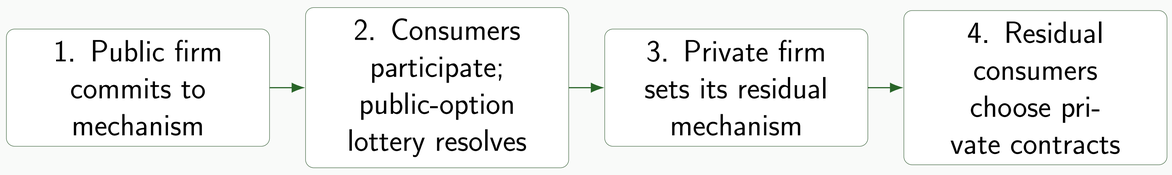
- Public and private provision coexist in markets such as housing and health care.
- With private market power, a public option changes not only who is served publicly, but also what a private seller offers to consumers who remain.
- Public provision therefore creates two effects:
  1. Direct reallocation: it trades off screening-distorted private provision against perhaps technologically less efficient public provision.
  2. Residual-market design: it selects the consumers left to the private seller and thereby reshapes its screening problem.
- How should the public offering be designed when these effects are taken into account? I show in a stylized model that the optimal public menu uses guaranteed low-quality provision and/or rationed high-quality provision.

Model

- A unit mass of consumers have WTP types $\theta \in \{L, H\}$, where $0 < L < H$ and $\Pr(\theta = L) = p$. Utility from quality $q \geq 0$ and transfer t is $\theta q - t$; the payoff from no trade is zero.
- The private and public per-consumer costs are $c_P(q)$ and $c_G(q)$. They satisfy $c_P(0) = c_G(0) = 0$, and on $q > 0$, they are strictly convex with

$$c_G(q) > c_P(q) > 0, \qquad c_G'(q) \geq c_P'(q) > 0, \qquad \lim_{q\to\infty} c_P'(q) = \lim_{q\to\infty} c_G'(q) = \infty.$$

- The public firm maximizes total surplus. The private firm maximizes profit.
- The public firm offers a mechanism where it can randomize whether a consumer gets a (positive-quality) public option; let x_θ denote type θ's probability of being allocated the public option, q_G^θ the quality conditional on public-option allocation, and t_G^θ the associated transfer.
- Consumers who do not get the public option enter the private market. Let (q_P^θ, t_P^θ) denote type θ's private allocation.



- Characterize the public firm's optimal PBE.

Benchmark: the public firm observes types

Suppose the public firm observes θ and can condition treatment directly on type.

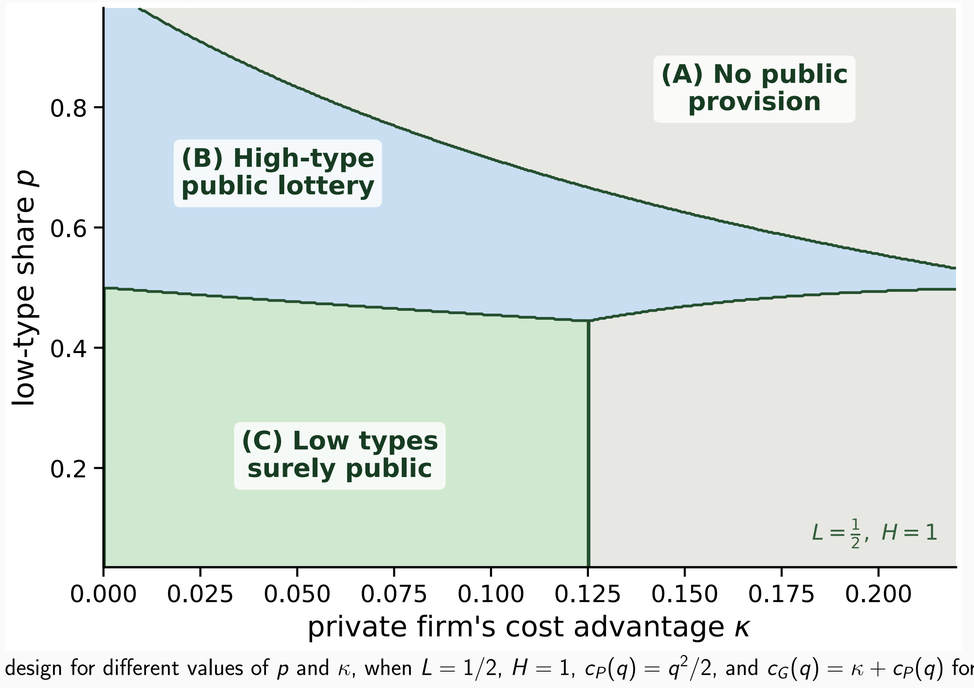
- Public qualities are then optimally efficient:

$$q_G^\theta = q_G^{\theta *} := \arg\max_{q\geq 0}\{\theta q - c_G(q)\}.$$

- The high type's private quality is efficient in private profit maximization. The low type's private quality depends on the composition of the residual market: moving more high types into the public option raises the private quality offered to low types.

Proposition 1 (observable types). There is an optimal public-option design that takes one of the following three forms:
(A) **No public provision:** $x_L = x_H = 0$.
(B) **High-type lottery:** $x_L = 0$, $x_H \in (0, 1)$, $q_G^H = q_G^{H*}$.
(C) **Guaranteed low-type provision:** $x_L = 1$, $x_H = 0$, $q_G^L = q_G^{L*}$.

- Low-type public provision is optimally either $x_L = 1$ or $x_L = 0$.
- If every low type is publicly served, $x_L = 1$, the private firm serves only high types. Public provision to high types would then create only an efficiency loss, so $x_H = 0$.
- High-only provision is never certain. Starting from $x_H = 1$, leaving an infinitesimal mass of high types private generates a first-order direct surplus gain but only a second-order low-type screening loss in the private market.



Optimal design for different values of p and κ, when $L = 1/2$, $H = 1$, $c_P(q) = q^2/2$, and $c_G(q) = \kappa + c_P(q)$ for $q > 0$.

Full model: types are private information

- Because transfers cancel from total surplus, the optimal design in the observable-type benchmark remains optimal whenever it can be implemented incentive-compatibly.

Lemma. Transfers satisfying public IC exist if and only if

$$x_H(q_G^H - q_P^L) \geq x_L(q_G^L - q_P^L).$$

- Design (C) fails IC when $q_G^L > 0$. Once low types leave, the private monopolist extracts all surplus from high types, who would therefore prefer the low-type public contract.
- When (C) solves the benchmark, the hidden-type optimum is therefore (A), (B), or a new hybrid design (C').
- (C') Hybrid provision: retain guaranteed provision to low types, but add a high-type lottery and distort public qualities from the efficient levels to restore IC.

Proposition 2 (hidden types). There is an optimal public-option design that takes one of the following three forms:
(A) **No public provision:** $x_L = x_H = 0$.
(B) **High-type lottery:** $x_L = 0$, $x_H \in (0, 1)$.
(C') **Hybrid provision:** $x_L = 1$ and $x_H \in (0, 1)$.

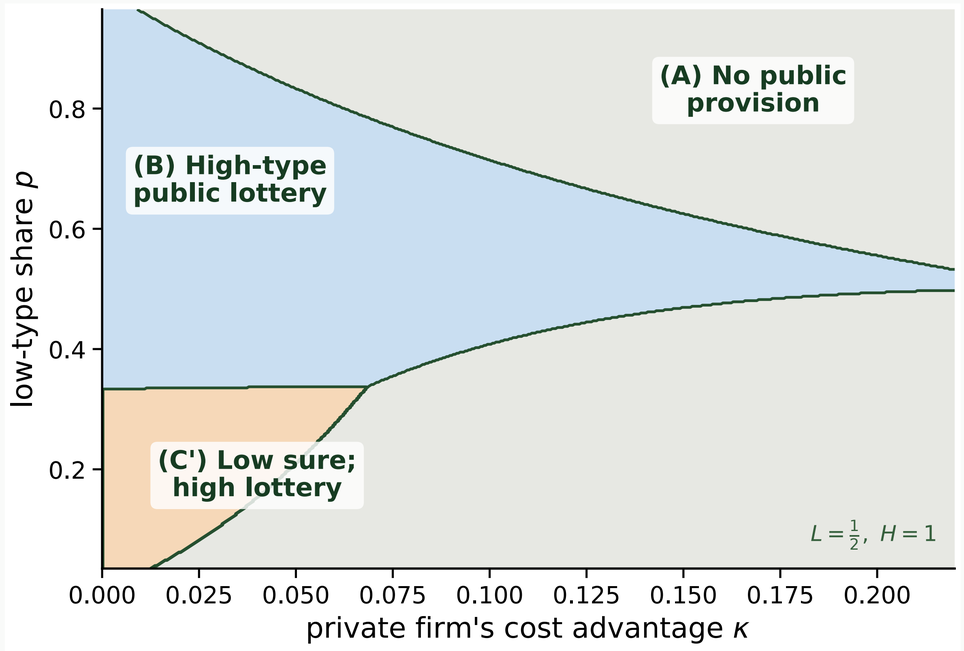


Illustration under the same specialization as the observable-type figure: $L = 1/2$, $H = 1$, $c_P(q) = q^2/2$, and $c_G(q) = \kappa + c_P(q)$ for $q > 0$.

Prior literature on the theory of mixed public–private markets

	Competitive private market	**Strategic private seller**
No public option design	Coate et al. (1994)	Mekonnen (2026)
Public option designed	Kang (2023); Kang–Watt (2026)	**This project**

Further directions

- Continuous WTP: how robust are the main insights of the binary case?
- Distributional objectives: allow different welfare weights on consumer surplus and private profit, possibly varying across consumer types.
- What if the public-option recipients are able to participate in the private market, making their public allocation an outside option for private screening?
- Alternative timing in the public and private options' designs

References

Coate, Johnson & Zeckhauser (1994), "Pecuniary Redistribution through In-Kind Programs," *JPubE*.
Kang (2023), "The Public Option and Optimal Redistribution."
Kang & Watt (2026), "Topping Up and Optimal Redistribution."
Mekonnen (2026), "Luck Out or Outpay? Competing with a Public Option."